Chapter 3

L^1 and L^∞ theory

Chapter 1 gave, in the framework of the general theory of optimal transportation based on duality methods, an existence result for the optimal transport map when the cost is of the form $c(x,y) = h(x-y)$ and h is strictly convex. In particular, using $c(x,y) = |x-y|^p$, this applies to the minimization problems

$$\min\left\{||\mathrm{T} - \mathrm{id}||_{L^p(\mu)} \ : \ \mathrm{T}_\#\mu = \nu\right\},$$

(expressed in Kantorovich terms as $\min\left\{||y-x||_{L^p(\gamma)} \ : \ \gamma \in \Pi(\mu,\nu)\right\}$), for $p \in]1,+\infty[$. We look in this chapter at the two limit cases $p = 1$ and $p = \infty$, which require additional techniques.

Then, the discussion section will go into two different directions: on the one hand the L^1 and L^∞ cases introduced and motivated the study of convex costs which could be non strictly-convex or infinite-valued somewhere, and the last developments on this topic will be debated in Section 3.3.1; on the other hand one could wonder what is the situation for $0 < p < 1$, and more generally for costs which are concave increasing functions of the distance, which will be the subject of Section 3.3.2.

3.1 The Monge case, with cost $|x-y|$

This section will prove that, if $\mu, \nu \in \mathscr{P}(\Omega)$, where $\Omega \subset \mathbb{R}^d$ is a compact domain, and $\mu \ll \mathscr{L}^d$, then there exists an optimal transport map T for the cost $|x-y|$ (the Euclidean distance in $\mathbb{R}^d$). The proof that we present is essentially due to Ambrosio [8], as a modification from that of Sudakov, [290]. In the introduction we gave references for the other alternative proofs of the existence in this case, and in the discussion section we will see their generalizations.

3.1.1 Duality for distance costs

Let us spend some words on the case where the cost $c(x,y)$ is actually a distance (thus satisfying the triangle inequality, vanishing if $x = y\ldots$). Since the cost is symmetric, we will avoid the distinction between the $\bar{c}$ and the c transform.

Proposition 3.1. *If $c : X \times X \to \mathbb{R}$ is a distance then a function $u : X \to \mathbb{R}$ is c-concave if and only if it is Lipschitz continuous with Lipschitz constant less than 1 w.r.t. the distance c. We will denote by Lip_1 the set of these functions. Moreover, for every $u \in \mathrm{Lip}_1$ we have $u^c = -u$.*

Proof. First take a c-concave function u. It can be written as

$$u(x) = \chi^c(x) = \inf_y \quad c(x,y) - \chi(y)$$

for some function $\chi : X \to \mathbb{R} \cup \{-\infty\}$. One can ignore the points y such that $\chi(y) = -\infty$ and, for the others, note that $x \mapsto c(x,y) - \chi(y)$ is a Lip_1 function, since $x \mapsto c(x,y)$ is Lip_1 as a consequence of the triangle inequality. Hence, $u \in \mathrm{Lip}_1$ since the infimum of a family of Lipschitz continuous functions with the same constant shares the same modulus of continuity.

Take now a function $u \in \mathrm{Lip}_1$. We claim that one can write

$$u(x) = \inf_y \quad c(x,y) + u(y).$$

Actually, it is clear that the infimum at the r.h.s. is not larger than $u(x)$, since one can always use $y = x$. On the other hand, $u \in \mathrm{Lip}_1$ implies $u(x) - u(y) \leq c(x,y)$, i.e. $u(x) \leq u(y) + c(x,y)$ for every y, and hence the infimum is not smaller than $u(x)$. This expression shows that $u = (-u)^c$ and that u is c-concave.

Applying this last formula to $-u$, which is also Lip_1, we get $u^c = -u$ and the last part of the claim follows. □ □

As a consequence, the duality formula, in the case of a distance cost function, gives

$$\min\left\{\int_{X\times X} c(x,y)\,\mathrm{d}\gamma,\ \gamma \in \Pi(\mu,\nu)\right\} = \max\left\{\int_X u\,\mathrm{d}(\mu-\nu)\ :\ u \in \mathrm{Lip}_1\right\}. \tag{3.1}$$

This also implies a useful property concerning the transport cost $\mathcal{T}_c$ when c is a distance, namely the fact that it satisfies the triangle inequality. We will see it again in Chapter 5, where transport costs $\mathcal{T}_c$ for $c(x,y) = |x-y|^p$ are used to define distances over $\mathcal{P}(\Omega)$. We want to stress it here since we will need it later on in this chapter.

Corollary 3.2. *If $c : X \times X \to \mathbb{R}$ is a distance, given $\mu, \nu, \rho \in \mathcal{P}(X)$ we have*

$$\mathcal{T}_c(\mu,\nu) \leq \mathcal{T}_c(\mu,\rho) + \mathcal{T}_c(\rho,\nu).$$

Proof. Take an arbitrary function $u \in \mathrm{Lip}_1$ and write

$$\int u\,\mathrm{d}(\mu-\nu) = \int u\,\mathrm{d}(\mu-\rho) + \int u\,\mathrm{d}(\rho-\nu) \leq \mathcal{T}_c(\mu,\rho) + \mathcal{T}_c(\rho,\nu).$$

The result follows by taking the supremum in u. □ □

We also note another useful property, typical of distances

Corollary 3.3. *If $c : X \times X \to \mathbb{R}$ is a distance and μ, ν are such that $\mathcal{T}_c(\mu,\nu) = 0$, then $\mu = \nu$*

Proof. Many proofs can be given of this simple fact. For instance, the duality formula gives $\int u\,\mathrm{d}(\mu-\nu)=0$ for every $u\in\mathrm{Lip}_1$, which is enough to guarantee $\mu=\nu$. Otherwise, one can simply note that $\mathscr{T}_c(\mu,\nu)=0$ implies the existence of $\gamma\in\Pi(\mu,\nu)$ such that $\int c\,\mathrm{d}\gamma=0$, i.e. $c(x,y)=0$ γ-a.e. Since c is a distance this means $x=y$ γ-a.e., which implies $\int\phi\,\mathrm{d}\mu=\int\phi(x)\,\mathrm{d}\gamma=\int\phi(y)\,\mathrm{d}\gamma=\int\phi\,\mathrm{d}\nu$ for every test function ϕ, and hence $\mu=\nu$. □ □

3.1.2 Secondary variational problem

As we already saw in the book-shifting example 2.15, the optimal γ for a distance cost like $|x-y|$ in $\mathbb{R}$ is not unique in general. We want now to select a special optimizer γ, so as to be able to prove that it is induced by a transport map. For simplicity, as we will use very soon some Euclidean features of the problem, in connection with the quadratic cost $|x-y|^2$, we switch back to a concrete presentation in a domain Ω of the Euclidian space $\mathbb{R}^d$. In particular, we will not use general distances, and the norm will only be the Euclidean norm.

Let us call $O(\mu,\nu)$ the set of optimal transport plans for the cost $|x-y|$. For notational simplicity we will also denote by K_p the functional associating to $\gamma\in\mathscr{P}(\Omega\times\Omega)$ the quantity $\int|x-y|^p\,\mathrm{d}\gamma$, and m_p its minimal value on $\Pi(\mu,\nu)$. In this language

$$O(\mu,\nu)=\mathrm{argmin}_{\gamma\in\Pi(\mu,\nu)}K_1(\gamma)=\{\gamma\in\Pi(\mu,\nu)\,:\,K_1(\gamma)\le m_1\}.$$

Note that $O(\mu,\nu)$ is a closed subset (w.r.t. the weak convergence of measures) of $\Gamma(\mu,\nu)$, which is compact. This is a general fact whenever we minimize a lower semi-continuous functional of γ.

From now on, we fix a Kantorovich potential u for the transport between μ and ν with cost $c(x,y)=|x-y|$, and we will use it as a tool to check optimality. Indeed, we have

$$\gamma\in O(\mu,\nu)\Leftrightarrow \mathrm{spt}(\gamma)\subset\{(x,y):u(x)-u(y)=|x-y|\}. \tag{3.2}$$

This is true because optimality implies $\int(u(x)-u(y))\,\mathrm{d}\gamma=\int|x-y|\,\mathrm{d}\gamma$ and the global inequality $u(x)-u(y)\le|x-y|$ gives equality γ-a.e. All these functions being continuous, the equality finally holds on the whole support. Viceversa, equality on the support allows to integrate it and prove that $K_1(\gamma)$ equals the value of the dual problem, which is the same of the primal, hence one gets optimality.

As we said, we want to select a special minimizer, somehow better than the others, and prove that it is actually induced by a map.

Let us consider the problem

$$\min\{K_2(\gamma)\,:\,\gamma\in O(\mu,\nu)\}.$$

This problem has a solution $\overline{\gamma}$ since K_2 is continuous for the weak convergence and $O(\mu,\nu)$ is compact for the same convergence. We do not know a priori about the uniqueness of such a minimizer. It is interesting to note that the solution of this problem may also be obtained as the limits of solutions γ_ε of the transport problem

$$\min\{K_1(\gamma)+\varepsilon K_2(\gamma),:\,\gamma\in\Pi(\mu,\nu)\},$$

but we will not exploit this fact here (its proof can be done in the spirit of Γ-convergence developments, as in Section 2.4).

The goal is now to characterize this plan $\overline{\gamma}$ and prove that it is induced by a transport map.

The fact that the condition $\gamma \in O(\mu,\nu)$ may be rewritten as a condition on the support of γ is really useful since it allows to state that $\overline{\gamma}$ also solves

$$\min\left\{\int c\,\mathrm{d}\gamma \,:\, \gamma\in\Pi(\mu,\nu)\right\},\ \text{ where } c(x,y)=\begin{cases}|x-y|^2 & \text{if } u(x)-u(y)=|x-y|\\ +\infty & \text{otherwise.}\end{cases}$$

Actually, minimizing this new cost implies being concentrated on the set where $u(x)-u(y)=|x-y|$ (i.e. belonging to $O(\mu,\nu)$) and minimizing the quadratic cost among those plans γ concentrated on the same set (i.e. among those $\gamma\in O(\mu,\nu)$).

Let us spend some words more in general on costs of the form

$$c(x,y)=\begin{cases}|x-y|^2 & \text{if } (x,y)\in A\\ +\infty & \text{otherwise,}\end{cases} \tag{3.3}$$

where A is a given closed subset of $\Omega\times\Omega$. First of all we note that such a cost is l.s.c. on $\Omega\times\Omega$.

Semi-continuity of the cost implies that optimal plans are concentrated on a set which is $c-CM$ (Theorem 1.43). What does it mean in such a case? c-cyclical monotonicity is a condition which imposes an inequality for every k, every σ and every family $(x_1,y_1),\dots,(x_k,y_k)$; here we will only need to use the condition for $k=2$, which gives the following result.

Lemma 3.4. *Suppose that $\Gamma\subset\Omega\times\Omega$ is $c-CM$ for the cost c defined in* (3.3). *Then*

$$(x_1,y_1),(x_2,y_2)\in\Gamma,(x_1,y_2),(x_2,y_1)\in A\Rightarrow(x_1-x_2)\cdot(y_1-y_2)\geq 0.$$

Proof. From the definition of c-cyclical monotonicity we have

$$(x_1,y_1),(x_2,y_2)\in\Gamma\Rightarrow c(x_1,y_1)+c(x_2,y_2)\leq c(x_1,y_2)+c(x_2,y_1).$$

For costs c of this form, this is only useful when both (x_1,y_2) and (x_2,y_1) belong to A (otherwise we have $+\infty$ at the right hand side of the inequality). If we also use the equivalence

$$|x_1-y_1|^2+|x_2-y_2|^2\leq|x_1-y_2|^2+|x_2-y_1|^2\Leftrightarrow(x_1-x_2)\cdot(y_1-y_2)\geq 0,$$

(which can be obtained just by expanding the squares), the claim is proven. □ □

3.1.3 Geometric properties of transport rays

Let us consider for a while the role played by the function u. We collect some properties.

Lemma 3.5. *If $x,y\in\Omega$ are such that $u(x)-u(y)=|x-y|$, then u is affine on the whole segment $[x,y]:=\{z=(1-t)x+ty,\ t\in[0,1]\}$.*

Proof. Take $z=(1-t)x+ty$. Just consider that the Lip_1 condition implies

$$u(x)-u(z)\le|x-z|=t|x-y],\quad u(z)-u(y)\le|z-y|=(1-t)|x-y].$$

Summing up the two inequalities we get $u(x)-u(y)\le|x-y|$, but the assumption is that this should be an equality. Hence we can infer that both inequalities are equalities, and in particular $u(z)=u(x)-t|x-y|$, i.e. $u((1-t)x+ty)=u(x)-t|x-y|$ is an affine function of t. □ □

Lemma 3.6. *If $z\in]x,y[:=\{z=(1-t)x+ty,\ t\in]0,1[\}$, for a pair of points $x\neq y$ such that $u(x)-u(y)=|x-y|$, then u is differentiable at z and $\nabla u(z)=e:=\frac{x-y}{|x-y|}$.*

Proof. First of all, let us give an estimate on the increment of u in directions orthogonal to e. Consider a unit vector h with $h\cdot e=0$ and a point $z'\in]x,y[$, and let t_0 be such that $z'\pm t_0e\in[x,y]$. Set $\delta:=u(z'+th)-u(z')$. By using $u\in\mathrm{Lip}_1$ we can say

$$u(z'+th)-u(z'-t_0e)=u(z'+th)-u(z')+u(z')-u(z'-t_0e)=\delta+t_0\le\sqrt{t^2+t_0^2}$$

as well as

$$u(z'+th)-u(z'+t_0e)=u(z'+th)-u(z')+u(z')-u(z'+t_0e)=\delta-t_0\le\sqrt{t^2+t_0^2}.$$

By raising these inequalities to power 2 we get

$$\delta^2+t_0^2\pm2t_0\delta\le t^2+t_0^2.$$

These two inequalities give $\pm2t_0\delta\le\delta^2\pm2t_0\delta\le t^2$, and hence $2t_0|\delta|\le t^2$, i.e. $|\delta|\le t^2/2t_0$.

Consider now a point $z\in]x,y[$ and a number $t_0<\min\{|z-x],|z-y|\}$. Any point z'' sufficiently close to z may be written as $z''=z'+th$ with h a unit vector orthogonal to e, $t\ll1$, $z'\in]x,y[$ such that $z'\pm t_0e\in[x,y]$. This allows to write

$$u(z'')-u(z)=u(z'+th)-u(z')+u(z')-u(z)=(z'-z)\cdot e+O(t^2)=(z''-z)\cdot e+O(t^2).$$

Using $O(t^2)=o(t)=o(|z''-z|)$, we get $\nabla u(z)=e$. □ □

Definition 3.7. We call transport ray any non-trivial (i.e. different from a singleton) segment $[x,y]$ such that $u(x)-u(y)=|x-y|$, which is maximal for the inclusion among segments of this form. The corresponding open segment $]x,y[$ is called the interior of the transport ray and x and y its boundary points. We call direction of a transport ray the unit vector $\frac{x-y}{|x-y|}$. We call $\mathrm{Trans}(u)$ the union of all non degenerate transport rays, $\mathrm{Trans}^{(b)}(u)$ the union of their boundary points and $\mathrm{Trans}(u)^{(i)}$ the union of their interiors. Moreover, let $\mathrm{Trans}^{(b+)}(u)$ be the set of upper boundary points of non-degenerate transport rays (i.e. those where u is minimal on the transport ray, say the points y in the definition $u(x)-u(y)=|x-y|$) and $\mathrm{Trans}^{(b-)}(u)$ the set of lower boundary points of non-degenerate transport rays (where u is maximal, i.e. the points x).

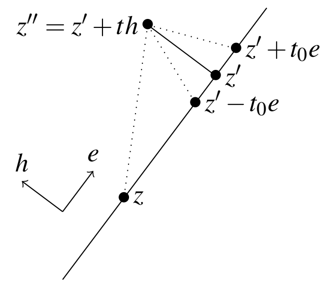

Figure 3.1: The points used in the proof of Lemma 3.6.

Corollary 3.8. *Two different transport rays can only meet at a point z which is a boundary point for both of them, and in such a case u is not differentiable at z. In particular, if one removes the negligible set* $\mathrm{Sing}(u)$ *of non-differentiability points of u, the transport rays are disjoint.*

Proof. First of all note that if a point z belongs to a transport ray with direction e, and u is differentiable at z, then necessarily we have $e = \nabla u(z)$. Indeed, close to z there are points of the form $z + te$ which belong to the same transport ray, for arbitrarily small values of t (we do not care about the sign of t), and on these points we can write

$$t = u(z+te) - u(z) = \nabla u(z) \cdot te + o(t).$$

This shows $\nabla u(z) \cdot e = 1$. Together with $|\nabla u(z)| \leq 1$, the only possibility is $\nabla u(z) = e$.

Now, suppose that two transport rays meet at a point z which is internal for both rays. In such a case u must have two different gradients at z, following both the directions of the rays, which is impossible (recall that the different rays meeting at one point must have different directions, since otherwise they are the same transport ray, by maximality).

Suppose that they meet at z, which is in the interior of a transport ray with direction e but is a boundary point for another ray, with direction $e' \neq e$. In such a case u should be differentiable at z and again $\nabla u(z)$ should take two distinct values, which is impossible.

Finally, suppose that the intersection point z is a boundary point for both segments, one with direction e and the other one with direction $e' \neq e$. In this case there is no contradiction if u is not differentiable at z. On the other hand, if one supposes that u is differentiable at z, then we get $\nabla u(z) = e = e'$, which is, again, a contradiction. □ □

We will see later on that we need to say something more on the direction of the transport rays.

From this section on we fix a transport plan $\overline{\gamma}$, optimal for the secondary variational problem, and we will try to prove that it is actually induced by a transport map. We use here that $\overline{\gamma}$ is actually concentrated on a set Γ which is c-CM (we recall that we are using the cost c defined as in (3.3)) and see how this interacts with transport rays. We can suppose $\Gamma \subset A = \{(x,y) : u(x) - u(y) = |x-y|\}$, since anyway $\overline{\gamma}$ must be concentrated on such a set, so as to have a finite value for $\int c \, d\gamma$. We want to say that $\overline{\gamma}$ behaves, on each transport ray, as the monotone increasing transport. More precisely, the following is true.

Lemma 3.9. *Suppose that x_1, x_2, y_1 and y_2 are all points of a transport ray $[x,y]$ and that $(x_1,y_1),(x_2,y_2) \in \Gamma$. Define an order relation on such a transport ray through $x \leq x' \Leftrightarrow u(x) \geq u(x')$. Then if $x_1 < x_2$ we also have $y_1 \leq y_2$.*

Proof. We already know that $x_1 \leq y_1$ and $x_2 \leq y_2$ (thanks to $\Gamma \subset A$). Hence, the only case to be considered is the case where we have $x_1 < x_2 \leq y_2 < y_1$. If we prove that this is not possible than we have proven the claim. And this is not possible due to the fat that Γ is c-CM, since on this transport ray, due to the order relationship we have supposed and to the behavior of u on such a segment, the condition $(x_1,y_2),(x_2,y_1) \in A$ is guaranteed. This implies $(x_1 - x_2) \cdot (y_1 - y_2) \geq 0$. But this is the scalar product of two vectors parallel to e, and on the segment this simply means that y_1 and y_2 must be ordered exactly as x_1 and x_2 are. □ □

From what we have seen in Chapter 2, we know that when s is a segment and $\Gamma' \subset s \times s$ is such that

$$(x_1,y_1),(x_2,y_2) \in \Gamma', \, x_1 < x_2 \Rightarrow y_1 \leq y_2$$

(for the order relation on s), then Γ' is contained in the graph of a monotone increasing multivalued function, which associates to every point either a point or a segment. Yet, the interiors of these segments being disjoint, there is at most a countable number of points where the image is not a singleton (see the proof of Lemma 2.8). This means that Γ' is contained in a graph over s, up to a countable number of points of s. We will apply this idea to $\Gamma' = \Gamma \cap (s \times s)$, where s is a transport ray.

If we combine what we obtained on every transport ray, we obtain:

Proposition 3.10. *The optimal transport plan $\overline{\gamma}$ is concentrated on a set Γ with the following properties:*

- *if $(x,y) \in \Gamma$, then*
  - *either $x \in \mathrm{Sing}(u)$ (but this set of points is Lebesgue-negligible),*
  - *or $x \notin \mathrm{Trans}(u)$, i.e. it does not belong to a non-degenerate transport ray (and in this case we necessarily have $y = x$, since otherwise $[x,y]$ would be contained in a non-degenerate transport ray),*
  - *or $x \in \mathrm{Trans}^{(b+)}(u) \setminus \mathrm{Sing}(u)$ (and in this case we necessarily have $y = x$, since x is contained in a unique transport ray s and it cannot have other images $y \in s$, due to the order relation $x \leq y$,*
  - *or $x \in \mathrm{Trans}(u) \setminus (\mathrm{Trans}^{(b+)}(u) \cup \mathrm{Sing}(u))$ and y belongs to the same transport ray s of x (which is unique);*
- *on each transport ray s, $\Gamma \cap (s \times s)$ is contained in the graph of a monotone increasing multivalued function;*

- *on each transport ray s, the set*

$$N_s = \{x \in s \setminus \mathrm{Trans}^{(b+)}(u) : \#(\{y : (x,y) \in \Gamma\}) > 1\}$$

is countable.

It is clear that $\overline{\gamma}$ is induced by a transport map if $\mu(\bigcup_s N_s) = 0$, i.e. if we can get rid of a countable number of points on every transport ray.

This could also be expressed in terms of disintegration of measures (if μ is absolutely continuous, then all the measures μ_s given by the disintegrations of μ along the rays s are atomless, see Box 2.2 in Section 2.3 for the notion of disintegration), but we will try to avoid such an argument for the sake of simplicity. The only point that we need is the following (*Property N*, for negligibility).

Definition 3.11. We say that Property N holds for a given Kantorovich potential u if, whenever $B \subset \Omega$ is such that

- $B \subset \mathrm{Trans}(u) \setminus \mathrm{Trans}^{(b+)}(u)$,
- $B \cap s$ is at most countable for every transport ray s,

then $|B| = 0$.

This property is not always satisfied by any disjoint family of segments in $\mathbb{R}^d$ and there is an example (by Alberti, Kirchheim and Preiss, later improved by Ambrosio and Pratelli, see [19]) where a disjoint family of segments contained in a cube is such that the collection of their middle points has positive measure. We will prove that the direction of the transport rays satisfy additional properties, which guarantee *property N*.

Please be aware that, as usual, we are ignoring here measurability issues of the transport map T that we are constructing: this map is obtained by gluing the monotone maps on every segment, but this should be done in a measurable way[1].

It appears that the main tool to prove Property N is the Lipschitz regularity of the directions of the transport rays (which is the same as the direction of ∇u).

Theorem 3.12. *Property N holds if ∇u is Lipschitz continuous or if there exists a countable family of sets E_h such that ∇u is Lipschitz continuous when restricted to each E_h and $|\mathrm{Trans}(u) - \setminus \bigcup_h E_h| = 0$.*

Proof. First, suppose that ∇u is Lipschitz. Consider all the hyperplanes parallel to $d-1$ coordinate axes and with rational coordinates on the last coordinate. Consider a set B in the definition of Property N. By definition, the points of B belong to non-degenerate transport rays, i.e. to segments with positive length. Hence, every point of B belongs to a transport ray that meets at least one of the above hyperplanes at exactly one point of its interior. Since these hyperplanes are a countable quantity, up to countable unions,

[1] Measurability could be proven, either by restricting to a σ-compact set Γ, or by considering the disintegrations μ_s and ν_s and using the fact that, on each s, T is the monotone map sending μ_s onto ν_s (and hence it inherits some measurability properties of the dependence of μ_s and ν_s w.r.t. s, which are guaranteed by abstract disintegration theorems).

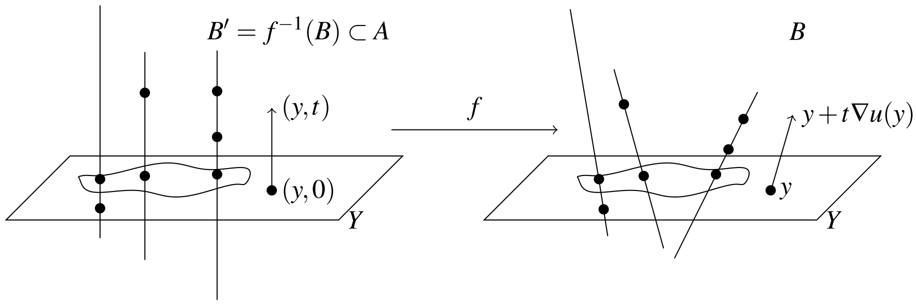

Figure 3.2: The map f in Theorem 3.12. The points of B and B' are only a countable number on each line, and represented by bullets.

we can suppose that B is included in a collection S_Y of transport rays all meeting the same hyperplane Y. Not only, we can also suppose that B does not contain any point which is a boundary point of two different transport rays, since we already know that those points are negligible. Now, let us fix such an hyperplane Y and let us consider a map $f : Y \times \mathbb{R} \to \mathbb{R}^d$ of the following form: for $y \in Y$ and $t \in \mathbb{R}$ the point $f(y,t)$ is defined as $y + t\nabla u(y)$. This map is well-defined and injective on a set $A \subset Y \times \mathbb{R}$ which is the one we are interested in. This set A is defined as those pairs (y,t) where y is in the interior of a transport ray of S_Y, hence u is differentiable at such a point, and $y + t\nabla u(y)$ belongs to the interior of the same transport ray. The map f is injective, since getting the same point as the image of (y,t) and of (y',t') would mean that two different transport rays cross at such a point. B is contained in the image of f by construction, so that f is a bijection between B and $B' := f^{-1}(B)$. The map f is also Lipschitz continuous, as a consequence of the Lipschitz behavior of ∇u. Note that B' is a subset of $Y \times \mathbb{R}$ containing at most countably many points on every line $\{y\} \times \mathbb{R}$. By Fubini's theorem, this implies $|B'| = 0$. Then we have also $|B| = |f(B')| \le \mathrm{Lip}(f)^d |B'|$, which implies $|B| = 0$.

It is clear that the property is also true when ∇u is not Lipschitz but is Lipschitz continuous on each set E_h of a partition covering almost all the points of $\mathrm{Trans}(u)$, since one can apply the same kind of arguments to all the sets $B \cap E_h$ and then use countable unions. □ □

We now need to prove that ∇u is countably Lipschitz.

Definition 3.13. A function $f : \Omega \to \mathbb{R}^d$ is said to be countably Lipschitz if there exist a countable family of sets E_h such that f is Lipschitz continuous on each E_h and $|\Omega \setminus \bigcup_h E_h| = 0$.

Note that "being Lipschitz continuous on a set E" or "being the restriction to E of a Lipschitz continuous function defined on the whole $\mathbb{R}^d$" are actually the same property, due to Lipschitz extension theorems. Indeed, whenever $E \subset X$ are metric spaces, every L-Lipschitz function $f : E \to \mathbb{R}$ can be extended to an $L-$Lipschitz function over X. This is known as Kirszbraun Theorem and can be easily obtained by using the functions f_k of the Memo Box 1.5, for $k \ge L$.

We want to prove that ∇u is countably Lipschitz. We will first prove that u coincides with some λ-convex or λ-concave functions on a sequence of sets covering almost everything. This requires a definition.

Definition 3.14. A function $f : \Omega \to \mathbb{R}$ is said to be λ-convex if $x \mapsto f(x) - \frac{\lambda}{2}|x|^2$ is convex, and λ-concave if $x \mapsto f(x) + \frac{\lambda}{2}|x|^2$ is concave. Note that the number λ is not required to be positive, so that λ-convex functions for $\lambda > 0$ are strictly convex, and, for $\lambda < 0$, they just have second derivatives (where they exist) which are bounded from below. Functions which are λ-convex (or λ-concave) for some values of λ are called *semi-convex* (or *semi-concave*).

Proposition 3.15. *There exist some sets E_h such that*

- $\bigcup_h E_h = \mathrm{Trans}(u) \setminus \mathrm{Trans}(u)^{(b+)}$,
- *on each set E_h the function u is the restriction of a λ-concave function, for a value λ depending on h.*

Proof. Let us define

$$E_h = \left\{ x \in \mathrm{Trans}(u) \,:\, \exists z \in \mathrm{Trans}(u) \text{ with } u(x) - u(z) = |x - z| > \frac{1}{h} \right\},$$

which is roughly speaking made of those points in the transport rays that are at least at a distance $\frac{1}{h}$ apart from the upper boundary point of the ray. It is clear that $E_h \subset \mathrm{Trans}(u) \setminus \mathrm{Trans}(u)^{(b+)}$. Moreover, we easily have $\bigcup_h E_h = \mathrm{Trans}(u) \setminus \mathrm{Trans}^{(b+)}(u)$.

Let us fix a function $c_h : \mathbb{R}^d \to \mathbb{R}$ with the following properties: $c_h \in C^2(\mathbb{R}^d)$, $|\nabla c_h| \leq 1$, $c_h(z) \geq |z|$ for all $z \in \mathbb{R}^d$, $c_h(z) = |z|$ for all $z \notin B(0, 1/h)$. Set $\lambda_h := -||D^2 c_h||_{L^\infty}$ (note that we can take this value to be of the order of $-\frac{1}{h}$). It is easy to check that, if $x \in E_h$, one has

$$u(x) = \inf_{y \in \mathbb{R}^d} |x - y| + u(y) \leq \inf_{y \in \mathbb{R}^d} c_h(x - y) + u(y) \leq \inf_{y \notin B(x, 1/h)} |x - y| + u(y) = u(x),$$

where the first inequality is a consequence of $|z| \leq c_h(z)$ and the second is due to the restriction to $y \notin B(x, 1/h)$. The last equality is justified by the definition of E_h. This implies that all the inequalities are actually equalities, and that $u(x) = u_h(x)$ for all $x \in E_h$, where

$$u_h(x) := \inf_{y \in \mathbb{R}^d} c_h(x - y) + u(y).$$

It is important to note that u_h is a λ_h-concave function.

Let us justify that u_h is λ_h-concave. With this choice of λ_h, c_h is obviously λ_h-concave. Consider

$$\begin{aligned} u_h(x) + \frac{\lambda_h}{2}|x|^2 &= \inf_{y \in \mathbb{R}^d} c_h(x - y) + \frac{\lambda_h}{2} 2|x|^2 + u(y) \\ &= \inf_{y \in \mathbb{R}^d} c_h(x - y) + \frac{\lambda_h}{2}|x - y|^2 + \lambda_h x \cdot y - \frac{\lambda_h}{2}|y|^2 + u(y). \end{aligned}$$

This last expression shows that $u_h(x)+\frac{\lambda_h}{2}|x|^2$ is concave in x, since it is expressed as an infimum of concave functions ($c_h(x-y)+\frac{\lambda_h}{2}|x-y|^2$ is concave, $\lambda_h x\cdot y$ is linear, and the other terms are constant in x). Hence $u_h(x)$ is λ_h-concave. □ □

The previous theorem allows to replace the function u with the functions u_h, which are more regular (since they are λ-concave, they share the same regularity of convex functions). Unfortunately, this is not enough yet, since convex functions in general are not even differentiable. A new countable decomposition is needed, and can be obtained from the following theorem that we do not prove.

Theorem 3.16. *If f is a convex function, then ∇f is countably Lipschitz.*

The proof of this theorem may be found in [16], in Theorem 5.34. It is also true when we replace ∇f with an arbitrary BV function, and this is the framework that one finds in [16].

Box 3.1. **– Memo** – BV functions in $\mathbb{R}^d$

On a domain $\Omega\subset\mathbb{R}^d$, an L^1 function $f:\Omega\to\mathbb{R}$ is said to be a BV function if its distributional derivatives are finite measures. This means that we require

$$\int_\Omega f\,(\nabla\cdot\xi)\,\mathrm{d}x\le C||\xi||_{L^\infty},$$

for every C^1_c vector field $\xi:\Omega\to\mathbb{R}^d$. The space BV is hence a wider class than the Sobolev spaces $W^{1,p}$, where the distributional derivatives are supposed to belong to L^p (i.e. they are absolutely continuous measures with integrability properties).

It happens that the distributional derivative ∇f cannot be any measure, but must be of the form $\nabla f=\nabla^a f(x)\,\mathrm{d}x+D^s f+D^j f$, where $\nabla^a f(x)\,\mathrm{d}x$ is the absolutely continuous part of ∇f (with respect to the Lebesgue measure), $D^j f$ is the so-called "jump part" and has a density with respect to $\mathscr{H}^{d-1}$ (it is concentrated on the set J_f of those points where there is an hyperplane such that the function f has two different limits on the two sides of the hyperplane, in a suitable measure-theoretical sense, and the density of $D^j u$ with respect to $\mathscr{H}^{d-1}_{|J_f}$ is exactly the difference of these two limit values times the direction of the normal vector to the hyperplane: $D^j f=(f^+-f^-)\mathbf{n}_{J_f}\cdot\mathscr{H}^{d-1}_{|J_f}$), and D^c is the so-called Cantor part, which is singular to the Lebesgue measure but vanishes on any $(d-1)$-dimensional set.

We denote by $BV_{loc}(\mathbb{R}^d)$ the space of functions which are locally BV in the sense that their derivatives are Radon measures, i.e. measures which are locally finite (in the definition with test functions ξ, we use $\xi\in C^1_c(\mathbb{R}^d)$ and the constant C may depend on the support of ξ). Vector-valued BV functions are just defined as functions $f=(f_1,\dots,f_k):\mathbb{R}^d\to\mathbb{R}^k$ which are componentwise BV, i.e. $f_i\in BV$ for each $i=1,\dots,k$. It is interesting that gradients $g=\nabla f$ of convex functions are always locally BV. This depends on the fact that the Jacobian matrix of the gradient of a convex function is indeed the Hessian of such a function, and is positive-definite. This means that the matrix-valued distribution $\nabla(\nabla f)=D^2 f$ is a positive distribution, and we know that positive distributions are necessarily positive measures (warning: this requires precise definitions when we work with matrix- and vector-valued functions).

BV functions satisfy several fine regularity properties almost everywhere for the Lebesgue or the $\mathscr{H}^{d-1}$-measure, and the reader may refer to [16] or [161]. In particular we cite the following (not at all easy) result, which implies Theorem 3.16.

Theorem - If $f \in BV_{loc}(\mathbb{R}^d)$, then f is countably Lipschitz.

As a consequence of Proposition 3.15 and Theorem 3.16 one has.

Proposition 3.17. *If u is a Kantorovich potential, then $\nabla u : \mathrm{Trans}(u) \to \mathbb{R}^d$ is countably Lipschitz.*

Proof. It is clear that the countably Lipschitz regularity of Theorem 3.16 is also true for the gradients of λ-convex and λ-concave functions. This means that this is true for ∇u_h and, by countable union, for ∇u. □ □

3.1.4 Existence and non-existence of an optimal transport map

From the analysis performed in the previous sections, we obtain the following result.

Theorem 3.18. *Under the usual assumption $\mu \ll \mathscr{L}^d$, the secondary variational problem admits a unique solution $\overline{\gamma}$, which is induced by a transport map* T*, monotone non-decreasing on every transport ray.*

Proof. Proposition 3.17, together with Theorem 3.12, guarantees that Property N holds. Hence, Proposition 3.10 may be applied to get $\overline{\gamma} = \gamma_{\mathrm{T}}$. The uniqueness follows in the usual way: if two plans $\overline{\gamma}' = \gamma_{\mathrm{T}'}$ and $\overline{\gamma}'' = \gamma_{\mathrm{T}''}$ optimize the secondary variational problem, the same should be true for $\frac{1}{2}\gamma_{\mathrm{T}'} + \frac{1}{2}\gamma_{\mathrm{T}''}$. Yet, for this measure to be induced by a map, it is necessary to have $\mathrm{T}' = \mathrm{T}''$ a.e. (see Remark 1.19). □ □

Definition 3.19. The optimal transport plan $\overline{\gamma}$ will be called *ray-monotone* transport plan, and the transport map which corresponds to it *ray-monotone* transport map.

Note that as a byproduct of this analysis we also obtain $|\mathrm{Trans}(u)^{(b)}| = 0$, since $\mathrm{Trans}(u)^{(b)}$ is a set meeting every transport ray in two points, and it is hence negligible by Property N. Yet we could not have proven it before, so unfortunately every strategy based on a decomposition of $\mathrm{Trans}(u)^{(i)}$ is not complete.

To complete this section, one could wonder whether the assumption $\mu \ll \mathscr{L}^d$ is really necessary for this existence result: would the condition "$\mu(A) = 0$ for every $(d-1)$-rectifiable set A" (or other conditions as in Section 1.3.1) be enough for the existence of an optimal map, as it happened for the quadratic case?

The answer is not, as it is shown in the following example.

Transport from a graph to a square Consider a continuous function $B : [0,1] \to \mathbb{R}$ with the following property: for every Lipschitz function $f : \mathbb{R} \to \mathbb{R}$, the sets $\{t \in [0,1] : B(t) = f(t)\}$ and $\{t \in [0,1] : f(B(t)) = t\}$ are Lebesgue-negligible. Then take $\mu = (\mathrm{id}, B)_\#(\mathscr{L}^1 \llcorner [0,1])$ and $\nu = \mathscr{L}^2 \llcorner Q$, where Q is the square $[0,1] \times [L, L+1]$ with $L = ||B||_{L^\infty}$.

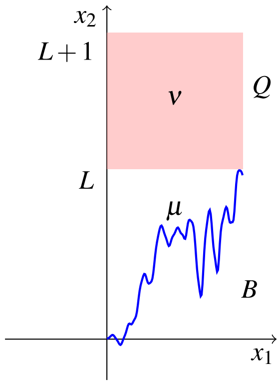

Figure 3.3: The measure μ concentrated on a graph, with ν on a square.

In practice, μ is concentrated on the graph of B and ν on a square above the graph. With this choice, the measure μ gives zero mass to any rectifiable curve of the form $\{(t, f(t))\}$ or $\{(f(s), s)\}$, for f Lipschitz. On the other hand, it is not absolutely continuous, as it is concentrated on a graph, which is $\mathscr{L}^2$-negligible. The goal is now to prove that no optimal transport map exists from μ to ν.

First consider that there is an obvious transport map S sending ν into μ. Just take $S(x_1,x_2) = (x_1, B(x_1))$ (i.e. each point moves vertically to the corresponding point on the graph of B). Obviously, S is not injective. The Monge cost of S is given by $\int_Q |x_2 - B(x_1)| \,\mathrm{d}x_1\,\mathrm{d}x_2 = \int_Q (x_2 - B(x_1))\,\mathrm{d}x_1\,\mathrm{d}x_2$. Moreover, the function $u(x_1,x_2) = -x_2$ is in Lip_1 and we have

$$\int_{\mathbb{R}^2} u\,\mathrm{d}(\mu-\nu) = -\int_0^1 B(x_1)\,\mathrm{d}x_1 + \int_Q x_2\,\mathrm{d}x_1\,\mathrm{d}x_2 = \int_Q |x_2 - B(x_1)|\,\mathrm{d}x_1\,\mathrm{d}x_2.$$

This proves that S is an optimal map from ν to μ and u is a Kantorovich potential (more precisely: u is a Kantorovich potential from μ to ν, and $-u$ from ν to μ). But this also proves that any optimal transport plan between μ and ν must be concentrated on the set $\{(x,y) : y_2 - x_2 = |x-y|\}$, i.e. on $\{(x,y) : y_1 = x_1, \text{ and } x_2 \le y_2\}$. In particular, the transport can only occur vertically, exactly as it happens for S. Yet, there is no transport map from μ to ν with this property: indeed, the image of $\mathrm{spt}(\mu)$ through such a map could only contain one point in each vertical segment of $\mathrm{spt}(\nu)$, and could not cover almost all Q.

To make the example complete[2], one only needs to produce such a function B. As the reader could imagine, the choice of the notation has been done on purpose: the typical trajectory of the Brownian motion satisfies this assumption. Indeed, it is known that the trajectories of a Brownian motion are almost surely never differentiable, and even never approximately differentiable (see the Box 3.3 in Section 3.3.2 for this notion, and see [223] for the behavior of the Brownian motion with this respect). By the

[2] Note that the construction is essentially the same as in the example provided in [200], for a different goal. The regularity degree is slightly different, and we decided to handle by hands "vertical" Lipschitz curves in order to make a self-contained presentation.

way, it is also proven in [26] that, almost surely, there is no set A with dimension bigger than $1/2$ on which B could be Hölder continuous with exponent larger than $1/2$. This prevents the possibility that B coincides with a Lipschitz function on a 1-dimensional set. We can even say that $\{t \in [0,1] : B(t) = f(t)\}$ has dimension less or equal than $1/2$.

For the other set (i.e. the points where $t = f(B(t))$), it is trickier. We can assume, by contradiction, that there exists a compact set K with $|K| > 0$ and $f \circ B = \mathrm{id}$ on K. Hence, the set $H = B(K)$ is also compact, and must have positive measure. Otherwise, if $|H| = 0$, $K \subset f(H)$ would be negligible too, because f is Lipschitz. We can assume that f is C^1, if we restrict H. Indeed, for every Lipschitz function f and every n there is a C^1 function $\widetilde{f}_n$ with $|\{f \neq \widetilde{f}_n\}| < \frac{1}{n}$ (see [161] for this fact). If we set $H_n = \{f = \widetilde{f}_n\}$ we have $|H \setminus \bigcup_n H_n| = 0$ and $|f(H \setminus \bigcup_n H_n)| = 0$; this proves $|f(\bigcup_n H_n)| = |\bigcup_n f(H_n)| > 0$ and at least for some n we must have $|H_n|, |f(H_n)| > 0$. Hence, we can suppose $f \in C^1$. Now, there are two possibilities: either $f' = 0$ on H, or there is a small interval where f is a diffeomorphism. In this second case we find $B = f^{-1}$ on a set of positive measure, which is a contradiction with the well-known properties of Brownian motions (since f^{-1} is Lipschitz, and even C^1). On the contrary, if $f' = 0$ on H then the area formula provides $|K| = 0$, which is also a contradiction.

3.1.5 Approximation issues for the ray-monotone optimal transport

The present section deals with some natural questions concerning the ray-monotone optimal transport plan $\overline{\gamma}$, and in particular approximation questions. Indeed, we have seen (Theorem 1.50) that for any weakly converging sequence $\gamma_n \rightharpoonup \gamma$, if γ_n is optimal for a continuous cost between its marginal, then so is γ. Yet, we know that for $c(x,y) = |x-y|$ the optimal transport plan is not in general unique, and for many applications one prefers to stick to the ray-monotone one. Hence, the question is how to select this special transport plan by approximation, when the measures and/or the cost vary.

For many applications (see in particular Chapter 4), it is interesting to approximate transport plans through transport maps sending a given measure to an atomic one. This is because this kind of transport maps is actually composed of different homotheties defined on a partition of the domain.

Here is a useful approximation lemma in the spirit of Γ-convergence developments (see [134], Section 2.4 and Box 4.6 in Section 4.4.2).

For fixed measures $\mu, \nu \in \mathscr{P}(\Omega)$, with $\mu \ll \mathscr{L}^d$, consider the following family of minimization problems (P_ε):

$$(P_\varepsilon) = \min \Big\{ \mathscr{T}_c((\pi_y)_\# \gamma, \nu) + \varepsilon K_1(\gamma) + \varepsilon^2 K_2(\gamma) + \varepsilon^{3d+3} \#((\pi_y)_\# \gamma) \ : \\ \gamma \in \mathscr{P}(\Omega \times \Omega),\ (\pi_x)_\# \gamma = \mu \Big\},$$

where $\mathscr{T}_c$ is the minimum value of the transport problem for the cost $c(x,y) = |x-y|$, $K_p(\gamma) = \int |x-y|^p \gamma(\mathrm{d}x, dy)$ for $p = 1,2$ and the symbol # denotes the cardinality of the support of a measure. Note that $K_1(\gamma) \neq \mathscr{T}_c((\pi_x)_\# \gamma, (\pi_y)_\# \gamma)$ in general.

Actually, the minimization of (P_ε) consists in looking for a transport plan with first marginal equal to μ satisfying the following criteria with decreasing degree of importance: the second marginal must be close to ν, the K_1 cost of transportation should be small, the K_2 as well, and, finally, the second marginal must be atomic with not too many atoms.

This minimization problem has obviously at least a solution (by the direct method, since Ω is compact). We denote by γ_ε such a solution and $\nu_\varepsilon := (\pi_y)_\# \gamma_\varepsilon$ its second marginal. It is straightforward that ν_ε is an atomic measure and that γ_ε is the (unique, since $\mu \ll \mathscr{L}^d$) solution of $\min\{K_1(\gamma)+\varepsilon K_2(\gamma) \,:\, \gamma \in \Pi(\mu,\nu_\varepsilon)\}$, which is an optimal transport problem from μ to ν_ε for the cost $|x-y|+\varepsilon|x-y|^2$. Set

$$\overline{\gamma} = \operatorname{argmin}\{K_2(\gamma) \,:\, \gamma \text{ is a } K_1\text{-optimal transport plan from } \mu \text{ to } \nu\}. \tag{3.4}$$

This transport plan $\overline{\gamma}$ is unique, and it is the unique ray-monotone optimal transport plan from μ to ν (this is a consequence of Section 3.1.4); note that the functional K_2 could have been replaced by any functional $\gamma \mapsto \int h(x-y)\,\mathrm{d}\gamma$ for a strictly convex function h).

Lemma 3.20. *As $\varepsilon \to 0$ we have $\nu_\varepsilon \rightharpoonup \nu$ and $\gamma_\varepsilon \rightharpoonup \overline{\gamma}$.*

Proof. It is sufficient to prove that any possible limit of subsequences coincides with ν or $\overline{\gamma}$, respectively. Let γ_0 be one such a limit and $\nu_0 = (\pi_y)_\#\gamma_0$ the limit of the corresponding subsequence of ν_ε. Consider a regular grid $G^n \subset \Omega$ composed of approximately Cn^d points (take $G^n = \frac{1}{n}\mathbb{Z}^d \cap \Omega$), and let p_n be any measurable map from Ω to G^n, with the property $|p_n(x)-x| \le 1/n$ (for instance, p_n can be the projection map, where it is well-defined). Set $\nu^n := (p_n)_\#\nu$ and note $\#\nu^n \le Cn^d$, as well as $\nu^n \rightharpoonup \nu$ and $\mathscr{T}_c(\nu^n,\nu) \le \frac{1}{n}$.

First step: $\nu_0 = \nu$. Take γ^n any transport plan from μ to ν^n. By optimality of γ_ε we have

$$\mathscr{T}_c(\nu_\varepsilon,\nu) \le \mathscr{T}_c(\nu^n,\nu) + \varepsilon K_1(\gamma^n) + \varepsilon^2 K_2(\gamma^n) + C\varepsilon^{3d+3}n^d.$$

Fix n, let ε go to 0 and get

$$\mathscr{T}_c(\nu_0,\nu) \le \mathscr{T}_c(\nu^n,\nu) \le \frac{1}{n}.$$

Then let $n \to \infty$ and get $\mathscr{T}_c(\nu_0,\nu) = 0$, which implies $\nu_0 = \nu$ (thanks to Corollary 3.3).

Second step: γ_0 is optimal for K_1 from μ to ν. Take any optimal transport plan γ^n (for the K_1 cost) from μ to ν^n (i.e. $K_1(\gamma^n) = \mathscr{T}_c(\mu,\nu^n)$). These plans converge to a certain optimal plan $\widetilde{\gamma}$ from μ to ν (i.e. $K_1(\widetilde{\gamma}) = \mathscr{T}_c(\mu,\nu)$). Then, by optimality of γ_ε, we have

$$\begin{aligned}\varepsilon K_1(\gamma_\varepsilon) &\le \mathscr{T}_c(\nu^n,\nu) + \varepsilon K_1(\gamma^n) + \varepsilon^2 K_2(\gamma^n) + C\varepsilon^{3d+3}n^d \\ &\le \frac{1}{n} + \varepsilon K_1(\gamma^n) + C\varepsilon^2 + C\varepsilon^{3d+3}n^d.\end{aligned}$$

Then take $n \approx \varepsilon^{-2}$ and divide by ε:

$$K_1(\gamma_\varepsilon) \le \varepsilon + K_1(\gamma^n) + C\varepsilon + C\varepsilon^{d+2}.$$

Passing to the limit we get

$$K_1(\gamma_0) \le K_1(\widetilde{\gamma}) = \mathscr{T}_c(\mu,\nu),$$

which implies that γ_0 is also optimal.

Third step: $\gamma_0 = \overline{\gamma}$. Take any optimal transport plan γ (for the cost K_1) from μ to ν. Set $\gamma^n = (\mathrm{id}\times p_n)_\#\gamma$, where $(\mathrm{id}\times p_n)(x,y) := (x, p_n(y))$. We have $(\pi_y)_\#\gamma^n = \nu^n$. Then we have

$$\mathscr{T}_c(\nu_\varepsilon,\nu) + \varepsilon K_1(\gamma_\varepsilon) + \varepsilon^2 K_2(\gamma_\varepsilon) \le \mathscr{T}_c(\nu^n,\nu) + \varepsilon K_1(\gamma^n) + \varepsilon^2 K_2(\gamma^n) + C\varepsilon^{3d+3}n^d.$$

Moreover, using the triangle inequality of Corollary 3.2, we infer

$$K_1(\gamma_\varepsilon) \ge \mathscr{T}_c(\mu,\nu_\varepsilon) \ge \mathscr{T}_c(\mu,\nu) - \mathscr{T}_c(\nu_\varepsilon,\nu).$$

We also have

$$K_1(\gamma^n) \le K_1(\gamma) + \int |p_n(y) - y|\,\mathrm{d}\gamma(x,y) \le K_1(\gamma) + \frac{1}{n} = \mathscr{T}_c(\mu,\nu) + \frac{1}{n}.$$

Hence we have

$$(1-\varepsilon)\mathscr{T}_c(\nu_\varepsilon,\nu) + \varepsilon\mathscr{T}_c(\mu,\nu) + \varepsilon^2 K_2(\gamma_\varepsilon) \le \frac{1}{n} + \varepsilon\mathscr{T}_c(\mu,\nu) + \frac{\varepsilon}{n} + \varepsilon^2 K_2(\gamma^n) + C\varepsilon^{3d+3}n^d.$$

Getting rid of the first term (which is positive) in the left hand side, simplifying $\varepsilon\mathscr{T}_c(\mu,\nu)$, and dividing by ε^2, we get

$$K_2(\gamma_\varepsilon) \le \frac{1+\varepsilon}{n\varepsilon^2} + K_2(\gamma^n) + C\varepsilon^{3d+1}n^d.$$

Here it is sufficient to take $n \approx \varepsilon^{-3}$ and pass to the limit to get

$$K_2(\gamma_0) \le K_2(\gamma),$$

which is the condition characterizing $\overline{\gamma}$ (K_2-optimality among plans which are K_1-minimizers). $\square$ $\square$

This approximation result will be useful in Chapter 4. It is mainly based on the fact (already mentioned in Section 3.1.2) that, when we minimize the transport cost $|x-y| + \varepsilon|x-y|^2$, we converge to the solution of the secondary variational problem, i.e. to the ray-monotone transport map. As we said, any kind of strictly convex perturbation should do the same job as the quadratic one.

Yet, there are other approximations that are as natural as this one, but are still open questions. We list two of them.

Open Problem (stability of the monotone transport): take γ_n to be the ray-monotone optimal transport plan for the cost $|x-y|$ between μ_n and ν_n. Suppose $\mu_n \rightharpoonup \mu, \nu_n \rightharpoonup \nu$ and $\gamma_n \rightharpoonup \gamma$. Is γ the ray-monotone optimal transport plan between μ and ν?

Open Problem (limit of $\sqrt{\varepsilon^2 + |x-y|^2}$): take γ_ε to be the optimal transport plan for the cost $\sqrt{\varepsilon^2 + |x-y|^2}$ between two given measures μ and ν. If $\mathrm{spt}(\mu)\cap\mathrm{spt}(\nu) =$

$\emptyset$, then γ_ε can be easily proven to converge to the ray-monotone optimal transport plan (because, at the first non-vanishing order in ε, the perturbation of the cost is of the form $|x-y|+\varepsilon^2/|x-y|+o(\varepsilon^2)$, and the function $h(t)=1/t$ is strictly convex on $\{t>0\}$). Is the same true if the measures have intersecting (or identical) supports as well?

This last approximation is very useful when proving (or trying to prove) regularity results (see [211]).

3.2 The supremal case, L^∞

We consider now a different problem: instead of minimizing

$$K_p(\gamma)=\int |x-y|^p \,\mathrm{d}\gamma,$$

we want to minimize the maximal displacement, i.e. its L^∞ norm. This problem has been first adressed in [123], but the proof that we present here is essentially taken from [198].

Let us define

$$\begin{aligned}K_\infty(\gamma) := ||x-y||_{L^\infty(\gamma)} &= \inf\{m\in\mathbb{R}\,:\,|x-y|\le m \text{ for } \gamma-a.e.(x,y)\}\\ &= \max\{|x-y|\,:\,(x,y)\in\operatorname{spt}(\gamma)\}\end{aligned}$$

(where the last equality, between an L^∞ norm and a maximum on the support, is justified by the continuity of the function $|x-y|$).

Lemma 3.21. *For every $\gamma\in\mathscr{P}(\Omega\times\Omega)$ the quantities $K_p(\gamma)^{\frac{1}{p}}$ increasingly converge to $K_\infty(\gamma)$ as $p\to+\infty$. In particular $K_\infty(\gamma)=\sup_{p\ge1}K_p(\gamma)^{\frac{1}{p}}$ and K_∞ is l.s.c. for the weak convergence in $\mathscr{P}(\Omega\times\Omega)$. Thus, it admits a minimizer over $\Pi(\mu,\nu)$, which is compact.*

Proof. It is well known that, on any finite measure space, the L^p norms converge to the L^∞ norm as $p\to\infty$, and we will not reprove it here. This may be applied to the function $(x,y)\mapsto|x-y|$ on $\Omega\times\Omega$, endowed with the measure γ, thus getting $K_p(\gamma)^{\frac{1}{p}}\to K_\infty(\gamma)$. It is important that this convergence is monotone here, and this is true when the measure has unit mass. In such a case, we have for $p<q$, using Hölder (or Jensen) inequality

$$\int |f|^p \,\mathrm{d}\gamma \le \left(\int |f|^q \,\mathrm{d}\gamma\right)^{p/q}\left(\int 1 \,\mathrm{d}\gamma\right)^{1-p/q} = \left(\int |f|^q \,\mathrm{d}\gamma\right)^{p/q},$$

for every $f\in L^q(\gamma)$. This implies, by taking the p-th root, $||f||_{L^p}\le||f||_{L^q}$. Applied to $f(x,y)=|x-y|$ this gives the desired monotonicity.

From this fact, we infer that K_∞ is the supremum of a family of functionals which are continuous for the weak convergence (since K_p is the integral of a bounded continuous function, Ω being compact, and taking the p-th root does not break continuity). As a supremum of continuous functionals, it is l.s.c. and the conclusion follows. □ □